

CS5340

Uncertainty Modeling in AI

Assoc. Prof. Lee Gim Hee

AY 2026/27

Semester 1

Course Information

Lecturer:

Dr. Lee Gim Hee

Department of Computer Science

Office: COM2-03-54

Email: gimhee.lee@comp.nus.edu.sg

Class Schedule:

Every Monday, 1830hrs – 2130hrs

Venue:

LT 17

Only In-person Lectures; **No Webcast**

Teaching Assistants

Stefan Lionar

Department of Computer Science

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Yeo Qi Xun

Department of Computer Science

Email: qixunyeo@u.nus.edu

Mode of Assessments

- The grades of this module is based on **60% CA + 40% final exam**:
 1. 4x **coding assignments** (15% each; individual work)
 2. 40% **final exam** (**conducted in-person**, one-page A4 cheat sheet allowed)

Logistics: Assignments

- We will use **Python** as the programming language for the assignments.
- Nonetheless, you can use any programming language of your choice.
- But the helper functions and our support will be given **only in Python**.
- Ask my TAs on all questions regarding the assignments.

Assignment Late Policy

- All assignments are **due at 2359hrs** of the dates specified on the module schedule.
- 25% of the total marks **will be deducted** for each day of late submission.
- Deduction of marks does not apply to the late submissions **with valid reasons**. Please email me your reasons to seek for approval.

Honor Code

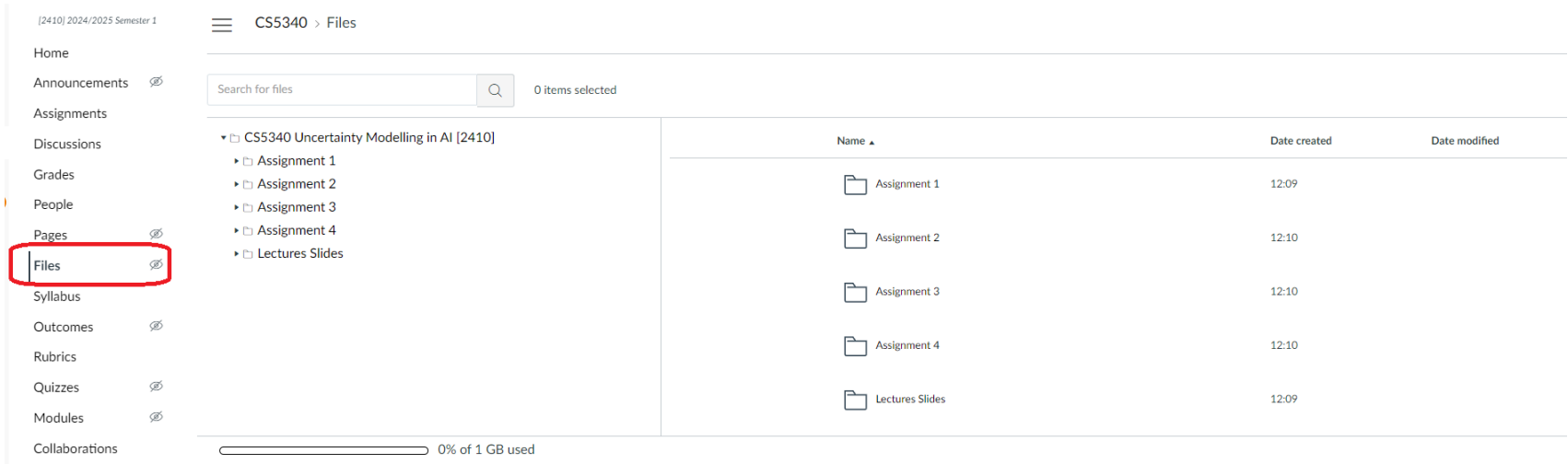
- **Assignments:** You may discuss and/or refer to online references, but **plagiarism** is strictly not allowed.
- Solutions directly from **Copilot**, **ChatGPT**, etc, are not allowed.
- **Violation of rules:** **Zero will be given**, and **disciplinary actions** that could lead to your expulsion from NUS will be taken!

Final Exam

- The final exam is to be conducted in-person at a **fixed date and time**.
- Please arrange your schedule, and make sure **you are present at NUS**.
- **Format (more details later):**
 1. Four questions
 2. 2-hour exam
 3. Students are allowed one-page A4 cheat sheet, written/typewritten on both sides.

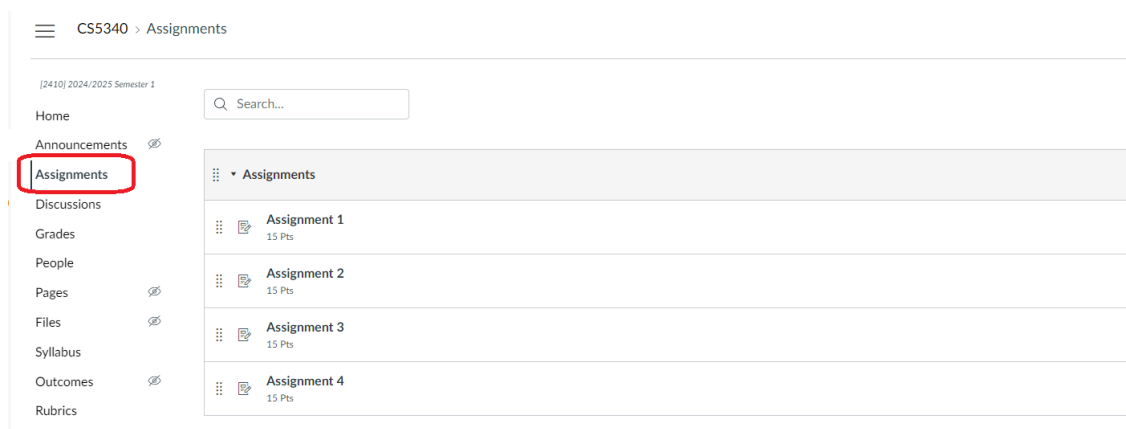
NUS Canvas

- We will use NUS Canvas for **accessing lecture materials** and **submission of assignments**.



The screenshot shows the NUS Canvas interface for the course CS5340. The left sidebar contains a list of navigation options: Home, Announcements, Assignments, Discussions, Grades, People, Pages, Files (highlighted with a red box), Syllabus, Outcomes, Rubrics, Quizzes, Modules, and Collaborations. The main content area is titled 'CS5340 > Files' and features a search bar and a table of files. The table has columns for Name, Date created, and Date modified. The files listed are Assignment 1, Assignment 2, Assignment 3, Assignment 4, and Lectures Slides, all created on 12:09. A progress bar at the bottom indicates 0% of 1 GB used.

Name	Date created	Date modified
Assignment 1	12:09	
Assignment 2	12:10	
Assignment 3	12:10	
Assignment 4	12:10	
Lectures Slides	12:09	



The screenshot shows the NUS Canvas interface for the course CS5340, specifically the Assignments page. The left sidebar contains a list of navigation options: Home, Announcements, Assignments (highlighted with a red box), Discussions, Grades, People, Pages, Files, Syllabus, Outcomes, Rubrics, Quizzes, Modules, and Collaborations. The main content area is titled 'CS5340 > Assignments' and features a search bar and a table of assignments. The table has columns for Name and Points. The assignments listed are Assignment 1 (15 Pts), Assignment 2 (15 Pts), Assignment 3 (15 Pts), and Assignment 4 (15 Pts).

Name	Points
Assignment 1	15 Pts
Assignment 2	15 Pts
Assignment 3	15 Pts
Assignment 4	15 Pts

Tutorials

- No formal tutorials.
- Two sets of exercise questions and solutions will be provided.
- I will go through some of the solutions during the lectures if time permits or during the last lecture.

Consultations

- Please send all questions to me **via email** and/or **Piazza Discussions (Preferred)**.
- To make sure your email gets my attention, use “[CS5340] xxx” as the title of your email.
- Subscribe to Piazza Discussions:
 1. <https://piazza.com/nus.edu.sg/fall2026/cs5340>
 2. Access code: CS5340-2026

Consultations

- Please send all your questions **on the assignments** to my TAs.
- Use the discussions forum in piazza!

Step 1:

Step 2:

The screenshot shows the Piazza interface for the course CS 5340. The top navigation bar includes 'Setup', 'Resources', 'Statistics', and 'Manage Class'. The 'Setup' tab is active, showing a list of folders: 'LIVE Q&A', 'Drafts', 'assignment_1', 'assignment_2', 'assignment_3', 'assignment_4', and 'lectures'. On the left sidebar, the 'New Post' button is highlighted. The main content area shows the 'Post Type' selection (Note is selected), 'Post To' selection (Entire Class is selected), and 'Select Folder(s)' dropdown (assignment_1, assignment_2, assignment_3, assignment_4, lectures). The 'Summary' field is highlighted, followed by the 'Details' section which includes a rich text editor. At the bottom, the 'Posting Options' are visible, and the 'Post My Note to CS 5340!' button is highlighted.

Step 3:

Step 4:

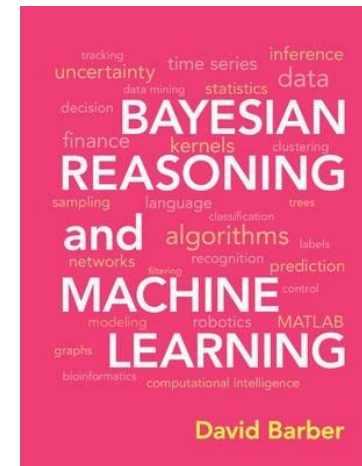
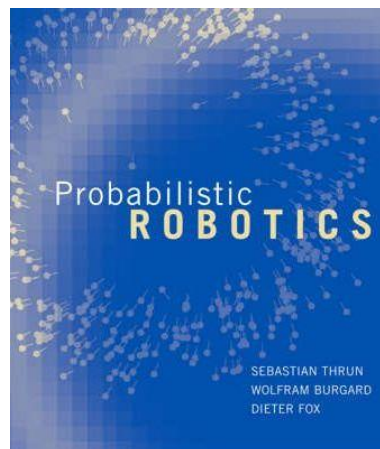
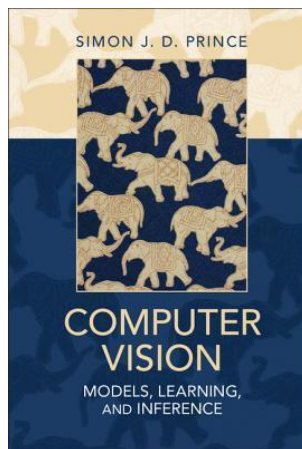
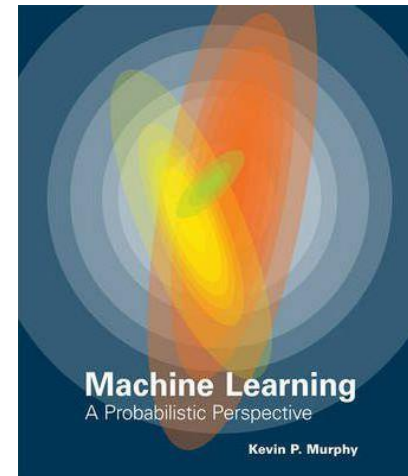
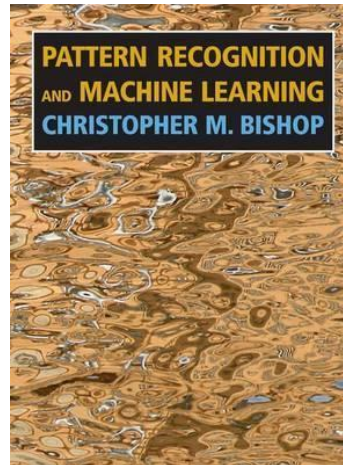
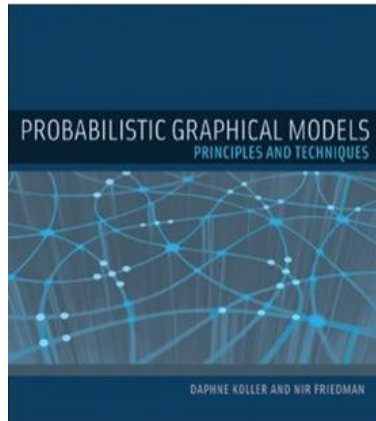
Step 5:

Course Schedule

Week	Date	Topic	Remarks
1	10 Aug	National Day Holiday	No Lecture
2	17 Aug	Introduction to probabilistic reasoning	Assignment 0: Python Numpy Tutorial (Ungraded)
	22 Aug	Bayesian networks (Directed graphical models)	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
3	24 Aug	Markov random Fields (Undirected graphical models)	
4	31 Aug	Variable elimination and belief propagation	Assignment 1: Belief propagation and maximal probability (15%) Assignment 1: Due (13 Sep, 2359 HRS)
	05 Sept	Factor graph and the junction tree algorithm	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
5	07 Sep	I will be traveling	No Lecture
6	14 Sep	Parameter learning with complete data	Assignment 2: Junction tree and parameter learning (15%) Assignment 2: Due (27 Sep, 2359 HRS)
-	21 Sep	Recess week	No Lecture
7	28 Sep	Mixture models and the EM algorithm	
8	05 Oct	Hidden Markov Models (HMM)	Assignment 3: Hidden Markov model (15%) Assignment 3: Due (18 Oct, 2359 HRS)
9	12 Oct	Monte Carlo inference (Sampling)	
10	19 Oct	Variational inference	Assignment 4: MCMC Sampling (15%) Assignment 4: Due (01 Nov, 2359 HRS)
11	26 Oct	Variational Autoencoder and Diffusion Models	
12	02 Nov	Graph-cut and alpha expansion	
13	09 Nov	Diwali/Deepavali	No Lecture

Final Exam: 21 Nov 2026

Recommended Readings (Not Compulsory)



Probabilistic Graphical Modeling

One of the most exciting advances in machine learning (AI, signal processing, coding, control, robotics, computer vision . . .) in the last decades.

Adapted from: “Probabilistic Graphical Modeling” Lectures NYU, David Sontag

Probabilistic Graphical Modeling

before deep learning

One of the most exciting advances in machine learning (AI, signal processing, coding, control, robotics, computer vision . . .) in the last decades.

Knowledge on PGM helps formulate some of the most important deep networks, e.g., deep generative models (Lecture 11) !

Adapted from: “Probabilistic Graphical Modeling” Lectures NYU, David Sontag

Probabilistic Graphical Modeling

How can we gain **global insight** based on **local observations**?

Adapted from: “Probabilistic Graphical Modeling” Lectures NYU, David Sontag

Probabilistic Graphical Modeling

How can we gain **global insight** based on **local observations**?

Example:

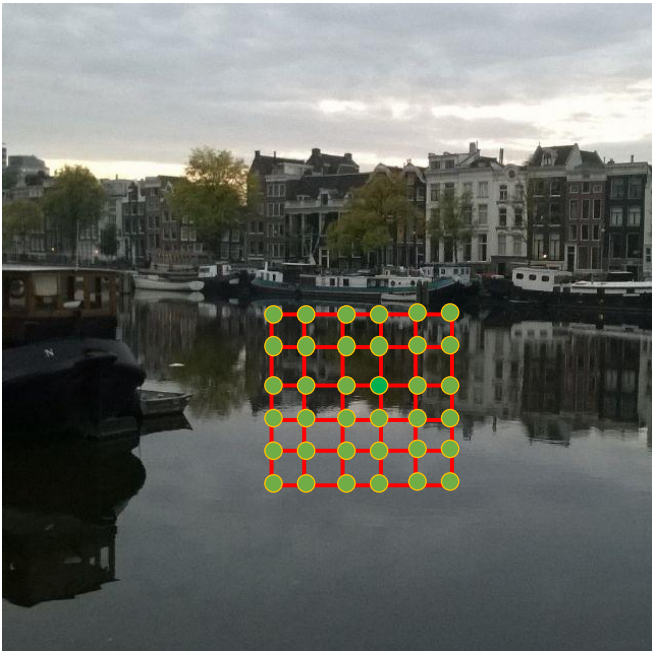


Photo Source:
G.H. Lee "Amsterdam"

Given: Local observations

- Each node takes 1-of-K labels and
- a smoothness prior, i.e, neighboring nodes linked by an edge should take the same label

We can find the label assignment of each pixel that is **globally consistent**!

Probabilistic Graphical Modeling

Key Ideas:

- **Represent** the world as a collection of random variables $X_1, \dots, X_N$ with joint distribution $p(X_1, \dots, X_N)$.
- **Learn** the distribution from data.
- Perform “**inference**” (compute conditional distributions $p(X_i \mid X_1 = x_1, \dots, X_N = x_N)$).

Adapted from: “Probabilistic Graphical Modeling” Lectures NYU, David Sontag

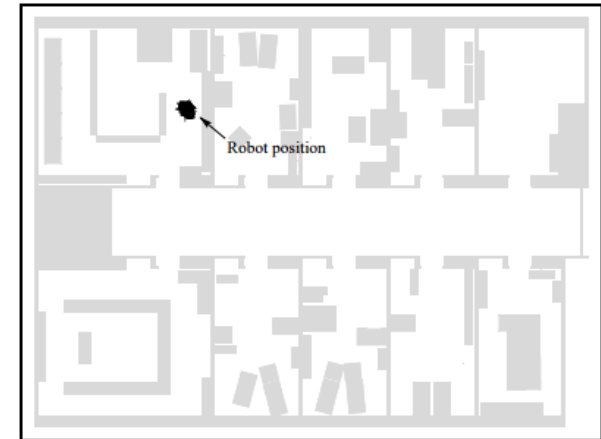
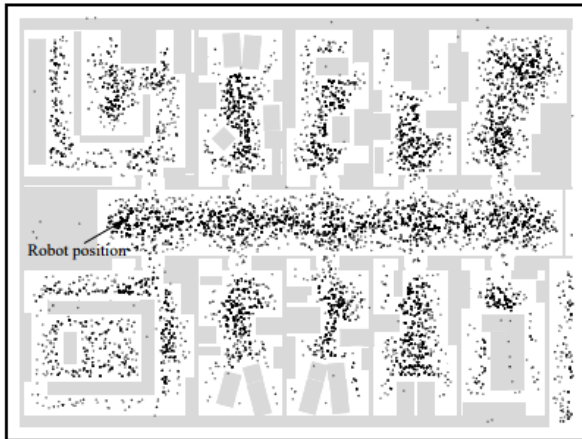
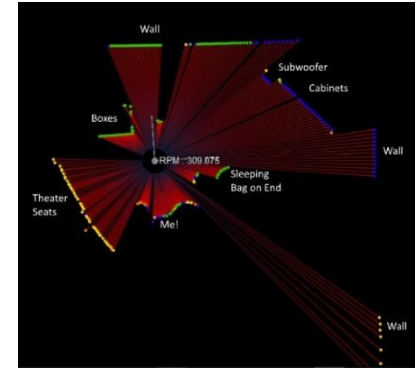
Reasoning Under Uncertainty

- As humans, we are continuously making **predictions under uncertainty**.
- Classical AI and ML research **ignored** this phenomena.
- Many of the ~~most recent~~ advances in technology are possible because of this **probabilistic approach**.

Adapted from: “Probabilistic Graphical Modeling” Lectures NYU, David Sontag

PGM: Applications

Markov Localization



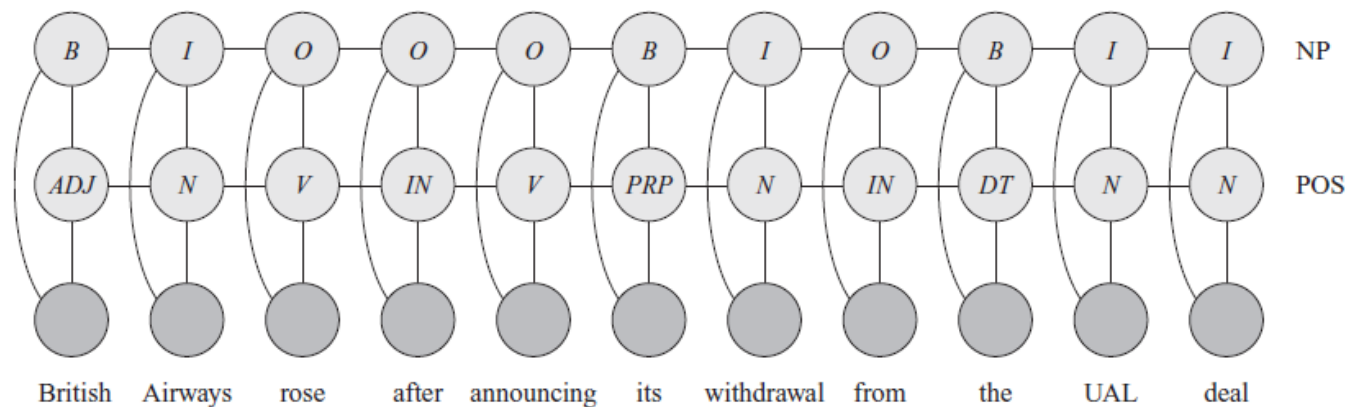
“Monte Carlo Localization for Mobile Robots”, Frank Dellaert et. al., ICRA 1999

PGM: Applications

Part of Speech Tagging

A. Big hungry **bears** are coming.

B. Your friend **bears** gifts.



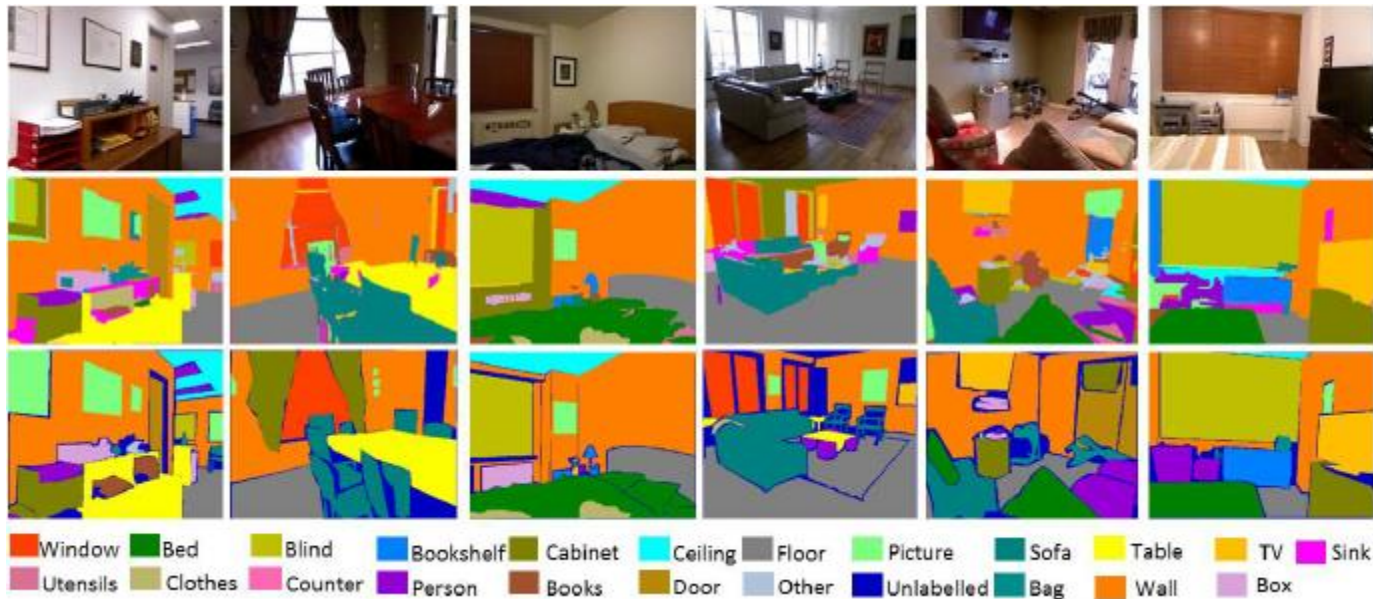
KEY

<i>B</i>	Begin noun phrase	<i>V</i>	Verb
<i>I</i>	Within noun phrase	<i>IN</i>	Preposition
<i>O</i>	Not a noun phrase	<i>PRP</i>	Possessive pronoun
<i>N</i>	Noun	<i>DT</i>	Determiner (e.g., a, an, the)
<i>ADJ</i>	Adjective		

D. Koller et. al. 2009

PGM: Applications

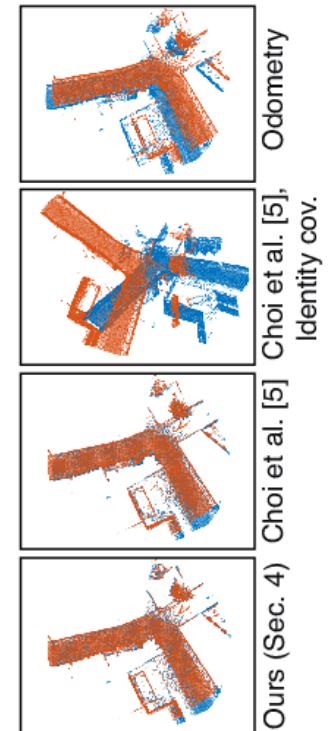
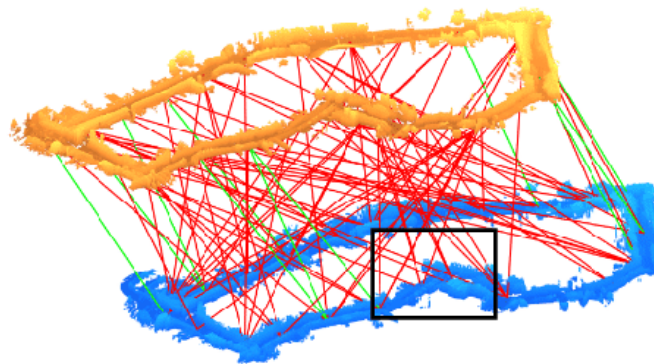
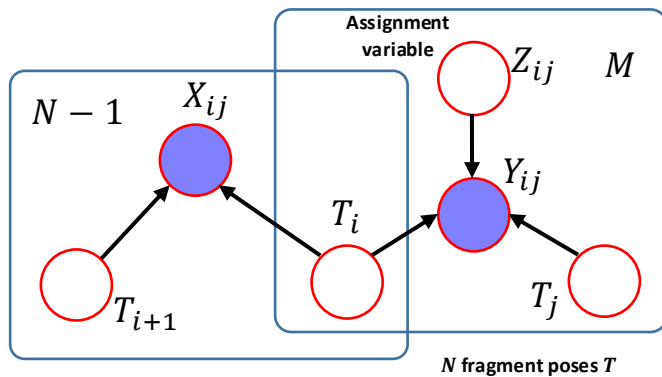
Scene Understanding



“Geometry Driven Semantic Labeling of Indoor Scenes”, Salman Hameed Khan et. Al. ECCV 2014

PGM: Applications

Robust 3D Reconstruction



Ziquan Lan, Zi Jian Yew, Gim Hee Lee, "Robust Point Cloud Based Reconstruction of Large-Scale Outdoor Scenes", CVPR 2019